QCA6410 HomePlug® AV MAC/PHY Transceiver

Data Sheet

MKG-16728 Ver. 5.0

March 16, 2012

Qualcomm Atheros Confidential

Confidential and Proprietary - Qualcomm Atheros, Inc.

Not to be used, copied, reproduced in whole or in part, nor its contents revealed in any manner to others without the express written permission of Qualcomm Atheros, Inc.

QUALCOMM is registered trademark of Qualcomm Incorporated. ATHEROS is a registered trademark of Qualcomm Atheros, Inc. All other registered and unregistered trademarks are the property of Qualcomm Incorporated, Qualcomm Atheros, Inc. or their respective owners and used with permission. Registered marks owned by Qualcomm Incorporated and Qualcomm Atheros, Inc. are registered in the United States of America and may be registered in other countries.

This technical data may be subject to U.S. and international export, re-export, or transfer (“export”) laws. Diversion contrary to U.S. and international law is strictly prohibited.

Qualcomm Atheros, Inc.
1700 Technology Drive
San Jose, CA 95110-1383
U.S.A.

Copyright © 2011-2012 QUALCOMM Incorporated.
All rights reserved.

Revision History

Revision	Date	Description
Ver. 1.0	August 2011	Preliminary QCA6410 Release
Ver. 2.0	November 2011	PL22 No-Build HDK Update Release
Ver. 3.0	December 2011	PL22 Alpha Release
Ver. 4.0	January 2012	Updated Ordering Information
Ver. 5.0	March 2012	PL22 No-Build HDK Update Release

Qualcomm Atheros Confidential

Contents

1 General Information 5
1.1 Description 5
1.2 Features 5
1.3 Applications 5
1.4 QCA6410 Functional Block Diagram 6
1.5 Conventions used in this data sheet 6

2 Pin Descriptions 7
2.1 QCA6410 Signals 8
2.1.1 QCA6410 Pin Assignments 8
2.1.2 Signal-to-Pin Relationships and Descriptions 11
2.2 System Introduction 14
2.2.1 System Overview 14
2.2.2 Integrated Functions 15
2.3 Zero Cross 16
2.4 QCA6410 Power On Configuration Strap Values and Defaults 17

3 System Architecture 20
3.1 Architecture Block Diagram 20
3.2 Host Interfaces 20
3.2.1 QCA6410 SPI Master Interfaces 20
3.2.2 QCA6410 Application Interface 21
3.2.3 GPIO 22

4 Functional Description 23
4.1 Overview 23
4.2 Power Management 23
4.2.1 Power Domains 23
4.2.2 Power Management Unit 23
4.3 Boot Options 25
4.4 Serial Interfaces 26

5 Electrical Characteristics 28
5.1 Absolute Maximum Ratings 28
5.2 Recommended Operating Conditions 29

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

5.3 QCA6410 Power Supply Requirements 29
5.4 DC Switching Thresholds 30
5.5 Crystal Specifications 30

6 Timing Constraints 31
6.1 System Clock and Reset Timing 31
6.1.1 QCA6410 Crystal Oscillator 31
6.1.2 Master Time Base Accuracy 32
6.1.3 External Clock Input 32
6.2 Zero Crossing Detector Timing 33
6.3 ISPI (Boot FLASH) Timing 33

7 Package Dimensions 35

8 Ordering Information 39

Qualcomm Atheros Confidential

1 General Information

1.1 Description

The QCA6410 is the latest addition to the Powerline Communications (PLC) portfolio of ICs from Qualcomm Atheros. It is fully compliant with the HomePlug-AV (HPAV) specification. The QCA6410 has been designed to meet specific low cost requirements for HomePlug-AV applications. The QCA6410 is interoperable with all HPAV networks and HomePlug Green PHY (HPGP) networks. The QCA6410 is a System-on-a-Chip (SoC) and offers lower power consumption, lower product complexity and lower BOM costs than previous ICs.

The QCA6410 is fully interoperable with HomePlug-AV and IEEE 1901 compliant products. The IC also supports Power Save Mode, which enables the IC to enter and exit a low power state on a scheduled or activity basis to maintain network synchronization.

An integrated switching power supply provides the 1.2 V core voltage. No external active components are required. Only inductors and capacitors are required for the power supply bypass circuit.

1.2 Features

- IEEE 1901 and HomePlug® AV compliant
- Co-existence with HomePlug® 1.0 Nodes
- Integrated 10/100 802.3az Ethernet PHY
- Interoperable with HomePlug AV specification and IEEE 1901 Standard
- Single chip solution with integrated analog front end and line driver
- Lower cost and power consumption than existing HPAV solutions for lower cost
- Host interfaces: GPIO and UART
- Boot from host or external FLASH memory
- Dedicated FLASH memory port (SPI master)
- Powered from single 3.3 VDC rail with integrated Power Management Unit
- RoHS Compliant: 8x8 mm 68 pin QFN

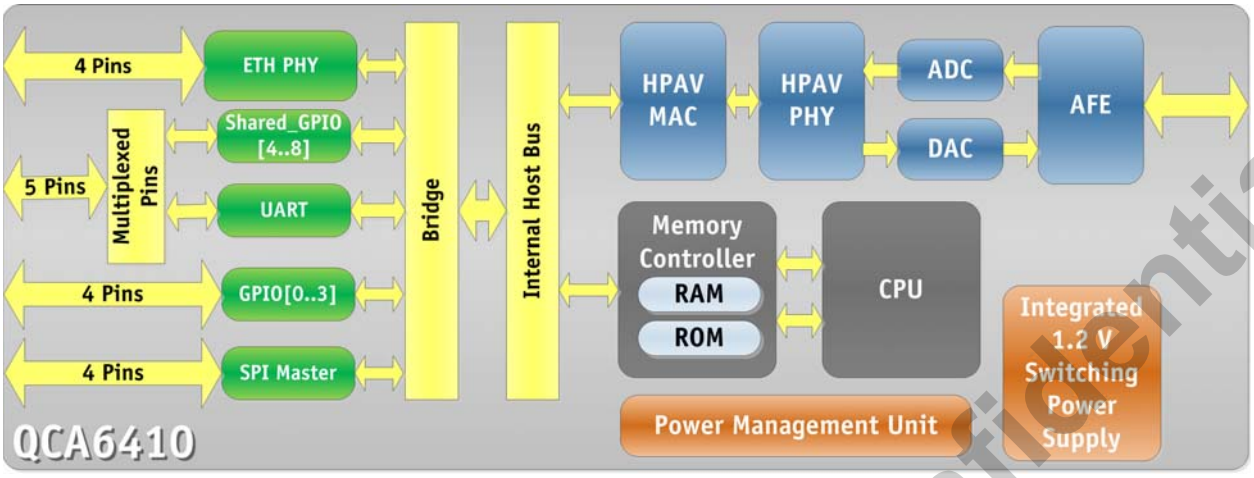
1.3 Applications

- HPAV bridge to Ethernet adapters
- PC files and applications sharing
- Printer and peripheral sharing
- Network and interactive online gaming

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

1.4 QCA6410 Functional Block Diagram



1.5 Conventions used in this data sheet

Unit or Abbreviation	Definition	Comment
k	Kilo-	1,000
M	Meg(a)-	1,000,000
b	bit(s)	
B	Byte(s)	
0x	Hexadecimal	
0b	Binary	
sps	Samples per second	
ps	Pico-second	
ns	Nano-second	
µs	Micro-second	
ms	Milli-second	
bps	Bits per second	
Mbps	Megabits per second	
µin	Micro-inches	
mil	Thousandth of an inch	0.001”
mm	Milli-meter	
Ω	Ohm	
mΩ	Milli-Ohm	
A	Ampere	
mA	Milli-Ampere	
V	Volts	
mV	Milli-Volts	0.001

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

2 Pin Descriptions

This section contains both a package pinout (see Figure 2-1, Table 2-1 and Table 2-2) and tabular listings of the signal descriptions.

The following nomenclature is used for signal names:

DNC	No connection should be made to this pin
_L	Low true signal
_P	At the end of the signal name, indicates the positive side of a differential signal
_N	At the end of the signal name indicates the negative side of a differential signal
RSVD	Reserved

The following nomenclature is used for signal types:

GND	A ground signal
I	Digital input signal
I/O	A digital bidirectional signal
IA	Analog input signal
IH	Input signals with weak internal pull-up, to prevent signals from floating when left open
IL	Input signals with weak internal pull-down, to prevent signals from floating when left open
O	A digital output signal
OA	An analog output signal
P	A power signal

Qualcomm Atheros Confidential

2.1 QCA6410 Signals

2.1.1 QCA6410 Pin Assignments

The QCA6410 package is a Pb-free, 8x8 mm 68-pin QFN

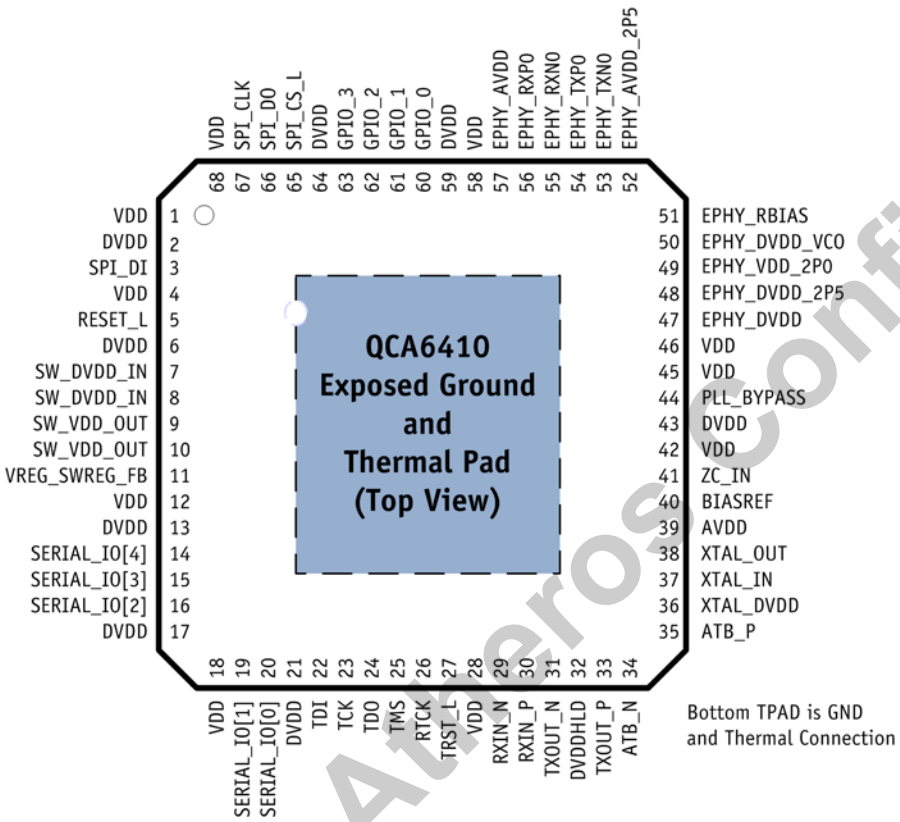


Figure 2-1 QCA6410 8x8 mm 68-Pin QFN

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-1 Signal-to-Pin Relationships and Descriptions

Pin	Symbol	Type	Interface	Description
1	VDD	P		1.2 V core voltage
2	DVDD	P		3.3 V
3	SPI_DI	I	SPI Master Bus	SPI Data In. The QCA6410 Master SPI_DI is used to transfer FW into the QCA6410. SPI_DI is latched on the rising edge of SPI_CLK. SPI Mode 3.
4	VDD	P		1.2 V core voltage
5	RESET_L	I	Reset Input	Power on Reset. Active-low Power-on-Reset input.
6	DVDD	P		3.3 V
7	SW_DVDD_IN	P	Switching Regulator	3.3 V input of internal Switching Regulator
8	SW_DVDD_IN	P	Switching Regulator	3.3 V input of internal Switching Regulator
9	SW_VDD_OUT	P	Switching Regulator	1.2 V output of internal Switching Regulator
10	SW_VDD_OUT	P	Switching Regulator	1.2 V output of internal Switching Regulator
11	VREG_SWREG_FB	P	Switching Regulator	Switching Regulator Feedback Pin. Connected to the regulated 1.2 V CORE supply bus.
12	VDD	P		1.2 V core voltage
13	DVDD	P		3.3 V
14	SERIAL_IO[4]	I/O	SERIAL_IO/GPIO	SERIAL I/O 4 - UART or Shared_GPIO_8
15	SERIAL_IO[3]	I/O	SERIAL_IO/GPIO	SERIAL I/O 3 - UART or Shared_GPIO_7
16	SERIAL_IO[2]	I/O	SERIAL_IO/GPIO	SERIAL I/O 2 - UART or Shared_GPIO_6
17	DVDD	P		3.3 V
18	VDD	P		1.2 V core voltage
19	SERIAL_IO[1]	I/O	SERIAL_IO/GPIO	SERIAL I/O 1 - UART or Shared_GPIO_5
20	SERIAL_IO[0]	I/O	SERIAL_IO/GPIO	SERIAL I/O 0 - UART or Shared_GPIO_4
21	DVDD	P		3.3 V
22	TDI	I	JTAG	10 K to GND for normal operation.
23	TCK	O	JTAG	10 K to GND for normal operation.
24	TDO	I/O	JTAG	3.3 K to GND for normal operation.
25	TMS	I	JTAG	10 K to GND for normal operation.
26	RTCK	O	JTAG	Leave unconnected for normal operation.
27	TRST_L	I	JTAG	10 K to GND for normal operation.
28	VDD	P		1.2 V core voltage
29	RXIN_N	IA	ADC/DAC	Differential In. Negative differential input from bandpass filter.
30	RXIN_P	IA	ADC/DAC	Differential In. Positive differential input from bandpass filter.
31	TXOUT_N	OA	ADC/DAC	Differential Out. Negative diff. output of the TX PGA

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-1 Signal-to-Pin Relationships and Descriptions (cont.)

Pin	Symbol	Type	Interface	Description
32	DVDDHLD	P	AFE	3.3 V Power for Analog Front End
33	TXOUT_P	OA	ADC/DAC	Differential Out. Positive diff. output of the TX PGA
34	ATB_N	OA	ADC/DAC	Make NO Connection. Analog Test pin. Reserved
35	ATB_P	OA	ADC/DAC	Make NO Connection. Analog Test pin. Reserved
36	XTAL_DVDD	P	ADC/DAC	3.3 V Analog
37	XTAL_IN	I	Crystal	Crystal Oscillator Inverter Input
38	XTAL_OUT	O	Crystal	Crystal Oscillator Inverter Output
39	AVDD	P	ADC/DAC	1.2 V Analog
40	BIASREF	P		External Bias Current from resistor
41	ZC_IN[1]	IA	Analog	Zero Cross Input. Minimum: 100 mVpp, AC coupled and maximum: 3.3 Vpp, AC coupled.
42	VDD	P		1.2 V core voltage
43	DVDD	P		3.3 V
44	PLL_BYPASS	I	Phase Lock Loop	Phase Lock Loop Bypass normally 10 KΩ to ground
45	VDD	P		1.2 V core voltage
46	VDD	P		1.2 V core voltage
47	EPHY_DVDD	P	Embedded E PHY	3.3 V for Embedded Ethernet PHY
48	EPHY_DVDD_2P5	P	Embedded E PHY	2.5 V for Embedded Ethernet PHY
49	EPHY_VDD_2P0	P	Embedded E PHY	2.0 V for Embedded Ethernet PHY
50	EPHY_DVDD_VCO	P	Embedded E PHY	1.2 V VCORE for PHY PLL
51	EPHY_RBIAS	P	Embedded E PHY	Embedded Ethernet PHY Reference BIAS
52	EPHY_AVDD_2P5	P	Embedded E PHY	2.5 V for Embedded Ethernet PHY
53	EPHY_TXN0	OA_N	Embedded E PHY	Embedded Ethernet PHY TX differential pair Neg.
54	EPHY_TXP0	OA_P	Embedded E PHY	Embedded Ethernet PHY TX differential pair Pos.
55	EPHY_RXN0	IA_N	Embedded E PHY	Embedded Ethernet PHY RX differential pair Neg.
56	EPHY_RXP0	IA_P	Embedded E PHY	Embedded Ethernet PHY RX differential pair Pos.
57	EPHY_AVDD	P	Embedded E PHY	1.2 V core for Embedded Ethernet PHY
58	VDD	P		1.2 V core voltage
59	DVDD	P		3.3 V
60	GPIO_0	I/O	GPIO	GPIO 0 Set configuration. Available after power on
61	GPIO_1	I/O	GPIO	GPIO 1 Set configuration. Available after power on
62	GPIO_2	I/O	GPIO	GPIO 2 Set configuration. Available after power on
63	GPIO_3	I/O	GPIO	GPIO 3 Set configuration. Available after power on
64	DVDD	P		3.3 V

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-1 Signal-to-Pin Relationships and Descriptions (cont.)

Pin	Symbol	Type	Interface	Description
65	SPI_CS_L	O	SPI Master Bus	SPI Chip Select. SPI_CS_N is the chip select used to enable the serial boot flash. This signal is tri-stated and held high with a weak internal pull-up while RESET_L is asserted.
66	SPI_DO	O	SPI Master Bus	SPI Data Out. SPI_DO is used to transfer serial data out of the QCA6410. SPI_DO is shifted out on the falling edge of SPI_CLK. This signal is tri-stated and held low with a weak internal pull-down while RESET_L is asserted.
67	SPI_CLK	O	SPI Master Bus	SPI Clock. SPI_CLK provides the SPI interface timing. Instructions, addresses, or data present at SPI_DI are latched on the rising edge of SPI_CLK. Data on SPI_DO changes after the falling edge of SPI_CLK. This signal is tri-stated and held low with a weak internal pull-down while RESET_L is asserted.
68	VDD	P		1.2 V core voltage
TPAD	GND	P	Power and Thermal	Power Ground and Thermal connection to PCB

1. If ZC_IN input is not used, connect ZC_IN to ground.

2.1.2 Signal-to-Pin Relationships and Descriptions

Table 2-2 Signal-to-Pin Relationships and Descriptions

Pin	Symbol	Type	Interface	Description
System Interface Signals				
5	RESET_L	I	Reset Input	Power on Reset. Active-low Power-on-Reset input.
44	PLL_BYPASS	I	Phase Lock Loop	Phase Lock Loop Bypass normally 10 KΩ to ground
Serial I/O See Table 3-2 and Table 3-3				
14	SERIAL_IO[4]	I/O	SERIAL_IO/GPIO	SERIAL I/O 4 - UART or Shared_GPIO_8
15	SERIAL_IO[3]	I/O	SERIAL_IO/GPIO	SERIAL I/O 3 - UART or Shared_GPIO_7
16	SERIAL_IO[2]	I/O	SERIAL_IO/GPIO	SERIAL I/O 2 - UART or Shared_GPIO_6
19	SERIAL_IO[1]	I/O	SERIAL_IO/GPIO	SERIAL I/O 1 - UART or Shared_GPIO_5
20	SERIAL_IO[0]	I/O	SERIAL_IO/GPIO	SERIAL I/O 0 - UART or Shared_GPIO_4
JTAG Port				
22	TDI	I	JTAG	10 K to GND for normal operation.
23	TCK	O	JTAG	10 K to GND for normal operation.
24	TDO	I/O	JTAG	3.3 K to GND for normal operation.
25	TMS	I	JTAG	10K to GND for normal operation.
26	RTCK	O	JTAG	Leave unconnected for normal operation.
27	TRST_L	I	JTAG	10 K to GND for normal operation.

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-2 Signal-to-Pin Relationships and Descriptions (cont.)

Pin	Symbol	Type	Interface	Description
Crystal Interface				
37	XTAL_IN	I	Crystal	Crystal Oscillator inverter Input
38	XTAL_OUT	O	Crystal	Crystal Oscillator Inverter Output
GPIO Bus Interfaces				
60	GPIO_0	I/O	GPIO	GPIO 0 Set configuration. Available after power on
61	GPIO_1	I/O	GPIO	GPIO 1 Set configuration. Available after power on
62	GPIO_2	I/O	GPIO	GPIO 2 Set configuration. Available after power on
63	GPIO_3	I/O	GPIO	GPIO 3 Set configuration. Available after power on
SPI Master Port for NVM Flash				
3	SPI_DI	I	SPI Master Bus	SPI Data In. The QCA6410 Master SPI_DI is used to transfer FW into the QCA6410. SPI_DI is latched on the rising edge of SPI_CLK. SPI Mode 3.
65	SPI_CS_L	O	SPI Master Bus	SPI Chip Select. SPI_CS_N is the chip select used to enable the serial boot flash. This signal is tri-stated and held high with a weak internal pull-up while RESET_L is asserted.
66	SPI_DO	O	SPI Master Bus	SPI Data Out. SPI_DO is used to transfer serial data out of the QCA6410. SPI_DO is shifted out on the falling edge of SPI_CLK. This signal is tri-stated and held low with a weak internal pull-down while RESET_L is asserted.
67	SPI_CLK	O	SPI Master Bus	SPI Clock. SPI_CLK provides the SPI interface timing. Instructions, addresses, or data present at SPI_DI are latched on the rising edge of SPI_CLK. Data on SPI_DO changes after the falling edge of SPI_CLK. This signal is tri-stated and held low with a weak internal pull-down while RESET_L is asserted.
Power Supply				
1	VDD	P		1.2 V core voltage
2	DVDD	P		3.3 V
4	VDD	P		1.2 V core voltage
6	DVDD	P		3.3 V
7	SW_DVDD_IN	P	Switching Regulator	3.3 V input of Switching Regulator
8	SW_DVDD_IN	P	Switching Regulator	3.3 V input of Switching Regulator
9	SW_VDD_OUT	P	Switching Regulator	1.2 V output of internal Switching Regulator
10	SW_VDD_OUT	P	Switching Regulator	1.2 V output of internal Switching Regulator
11	VREG_SWREG_FB	P	Switching Regulator	Switching Regulator Feedback Pin. Connected to the regulated 1.2 V supply BUS.
12	VDD	P		1.2 V core voltage
13	DVDD	P		3.3 V

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 2-2 Signal-to-Pin Relationships and Descriptions (cont.)

Pin	Symbol	Type	Interface	Description
17	DVDD	P		3.3 V
18	VDD	P		1.2 V core voltage
21	DVDD	P		3.3 V
28	VDD	P		1.2 V core voltage
32	DVDDHLD	P	AFE	3.3 V Power for Analog Front End
36	XTAL_DVDD	P	ADC/DAC	3.3 V Analog
39	AVDD	P	ADC/DAC	1.2 V Analog
40	BIASREF	P		External Bias Current from resistor
42	VDD	P		1.2 V core voltage
43	DVDD	P		3.3 V
45	VDD	P		1.2 V core voltage
46	VDD	P		1.2 V core voltage
47	EPHY_DVDD	P	Embedded E PHY	3.3 V for Embedded Ethernet PHY
48	EPHY_DVDD_2P5	P	Embedded E PHY	2.5 V for Embedded Ethernet PHY
49	EPHY_VDD_2P0	P	Embedded E PHY	2.0 V for Embedded Ethernet PHY
50	EPHY_DVDD_VCO	P	Embedded E PHY	1.2 V VCORE for PHY PLL
51	EPHY_RBIAS	P	Embedded E PHY	Embedded Ethernet PHY Reference BIAS
52	EPHY_AVDD_2P5	P	Embedded E PHY	2.5 V for Embedded Ethernet PHY
57	EPHY_AVDD	P	Embedded E PHY	1.2 V core for Embedded Ethernet PHY
58	VDD	P		1.2 V core voltage
59	DVDD	P		3.3 V
64	DVDD	P		3.3 V
68	VDD	P		1.2 V core voltage
TPAD	GND	P	Ground	Power Ground and Thermal connection to PCB
Analog Signals				
29	RXIN_N	IA	ADC/DAC	Differential In. Negative differential input from bandpass filter.
30	RXIN_P	IA	ADC/DAC	Differential In. Positive differential input from bandpass filter.
31	TXOUT_N	OA	ADC/DAC	Differential Out. Negative diff. output of the TX PGA
33	TXOUT_P	OA	ADC/DAC	Differential Out. Positive diff. output of the TX PGA
34	ATB_N	OA	ADC/DAC	Make NO Connection. Analog Test pin. Reserved
35	ATB_P	OA	ADC/DAC	Make NO Connection. Analog Test pin. Reserved
41	ZC_IN[1]	IA	Analog	Zero Cross Input. Minimum: 100 mVpp, AC coupled and maximum: 3.3 Vpp, AC coupled.
44	PLL_BYPASS	I	Phase Lock Loop	Phase Lock Loop Bypass normally 10 KΩ to ground

Qualcomm Atheros Confidential

Table 2-2 Signal-to-Pin Relationships and Descriptions (cont.)

Pin	Symbol	Type	Interface	Description
53	EPHY_TXN0	OA_N	Embedded E PHY	Embedded Ethernet PHY TX differential pair Neg.
54	EPHY_TXP0	OA_P	Embedded E PHY	Embedded Ethernet PHY TX differential pair Pos.
55	EPHY_RXN0	IA_N	Embedded E PHY	Embedded Ethernet PHY RX differential pair Neg.
56	EPHY_RXP0	IA_P	Embedded E PHY	Embedded Ethernet PHY RX differential pair Pos.

1. If ZC_IN input is not used, connect ZC_IN to ground.

2.2 System Introduction

The QCA6410 is designed to support HomePlug-AV applications. This document describes the details of the Atheros QCA6410 System on Chip (SoC) Integrated Circuit (IC).

2.2.1 System Overview

The QCA6410 is a complete system on a chip providing baseband Powerline Communication (PLC) processor/transceiver with serial and parallel data Input and Output (I/O). The key objectives are:

- Highly integrated IC for HPAV applications
- Integrated 10/100 802.3az Ethernet PHY
- Small, low-cost 68-QFN Package
- Compatible with full HomePlug (HP) 2 MHz to 30 MHz PLC channel
- Co-existence with HomePlug® 1.0 Nodes
- General Purpose single bit I/O - GPIO
- Serial data I/O supported by UART
- UART for Host Port interface
- Integrated ARM CPU & integrated memory for low pin count
- Integrated 1.2 V switching regulator for single 3.3 V supply operation
- Integrated Analog Front End for short distance communications
- IEEE1901 and HPAV interoperable
- 200 Mbps Powerline PHY Rate Max.

Qualcomm Atheros Confidential

2.2.2 Integrated Functions

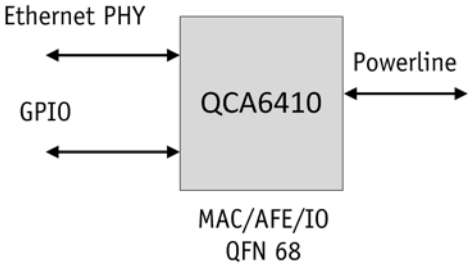


Figure 2-2 Basic Block Diagram

Configuration straps are read at power on and determine the function of various input and output pin options.

The following QCA6410 diagram Figure 2-2 shows some details of the internal functional blocks. The QCA6410 Analog Front End (AFE) includes the A/D and D/A converters, the analog receive amplifier, analog transmit amplifier, and analog line driver that supports the HomePlug-AV standard. An integrated analog zero crossing detector is included.

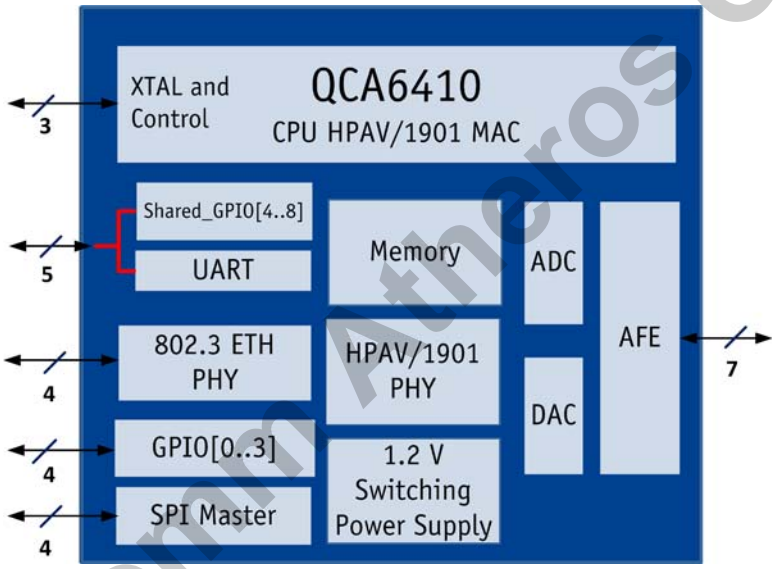


Figure 2-3 QCA6410 Internal Block Diagram

An external Powerline coupler passes HomePlug AV OFDM signals to and from the Powerline while providing isolation from hazardous Powerline voltages. The QCA6410 PHY is compliant with the IEEE 1901 Draft Standard with full support for the Inter System Protocol (ISP) including both in-home and access requirements up to the QCA6410 maximum data rate.

2.2.2.1 Core Voltage Power Supply

The QCA6410 includes an integrated on-board core voltage 1.2 V Power Supply requiring no external active components. Inductors and capacitors are required for the Pi network bypass circuit. See the QCA6410 Reference Design for component values and placement on the PCB. Multiple vias (four or more) are required on all through board connections delivering core voltage.

Qualcomm Atheros Confidential

2.2.2.2 Additional Internal Functions

Within the QCA6410, the HomePlug-AV MAC functions are executed in firmware running on the embedded ARM926 CPU, supported by DMA hardware, and on-chip SRAM for run-time code and data-store. The HomePlug-AV MAC firmware running on the CPU oversees operation of the integrated HomePlug-AV PHY via an interface that carries control/status information as well as transmits and receives data packets.

General-purpose I/O pins (GPIO) are available for application I/O and to drive LEDs directly to indicate PLC link status, as well as to indicate user reset or network-attach events. A dedicated QCA6410 IC sub-system is responsible for power-management functions.

An on-board PLL and on-chip crystal oscillator are used to generate the required system clocks. Various loop-back modes support manufacturing test.

2.3 Zero Cross

The QCA6410 has an analog amplifier circuit that detects when the 50 Hz or 60 Hz AC powerline voltage crosses through zero volts. This input pin is self biased so it is AC coupled by a capacitor. This input only requires a small AC waveform of about 100 mVpp. This small analog voltage requirement simplifies the external Zero Cross circuit. This greatly reduces power dissipation, lowers resulting heat and cost.

This temporal information synchronizes the channel adaptation of the QCA6410 IC to any line cycle periodic noise that may be present on the powerline. This synchronization information is used by software algorithms to optimize HP Powerline Communications in high noise conditions.

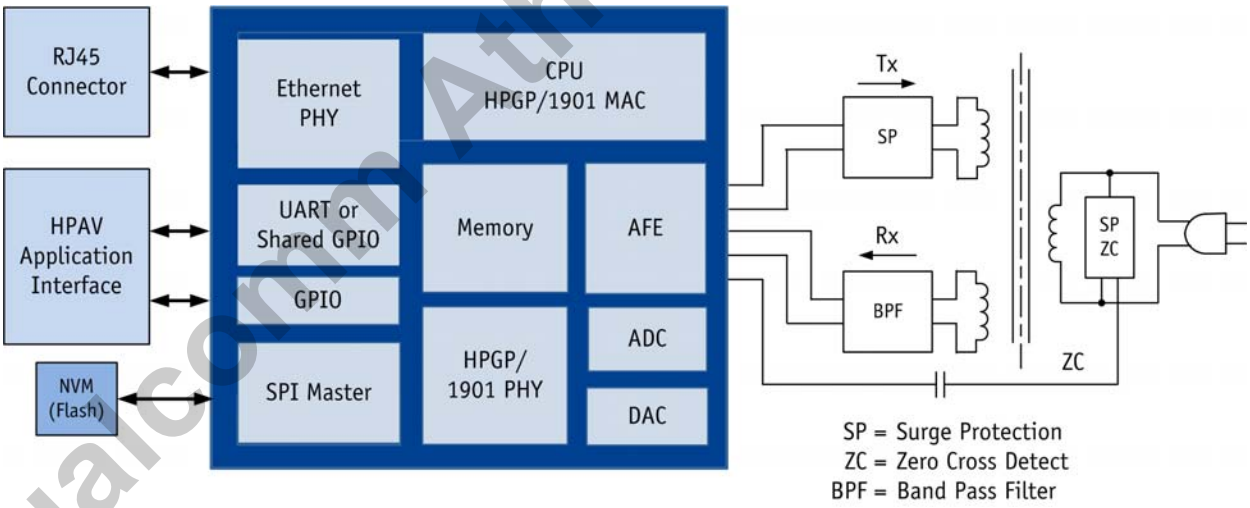


Figure 2-4 QCA6410 System Block Diagram

Qualcomm Atheros Confidential

2.4 QCA6410 Power On Configuration Strap Values and Defaults

The following table shows GPIO pins that are read at power on to configure the operation of the QCA6410. The default, internal pull-up and pull-down resistor is shown in the right hand column. It is good design practice to set strap pins high or low by placing a 3 K to 10 K resistor tied to ground or to the 3.3 V power supply. This makes testing and changing modes for test easy.

After the Reset signal goes high several GIPIO pins are used by default by the FW program or can be used by the designer for general purpose I/O functions.

Table 2-3 Configurative Strap Functions and Internal Defaults

Strap Function	Pin	QCA6410 Pin Name	Function Description	Internal Pull-Up/Down
Boot from NVM	60	GPIO_0	High (1) indicates boot code loads from NVM flash device on the Master SPI bus port.	PU (1)
No Function at Boot. User I/O after Boot	61	GPIO_1	Can be an output for TX/Rx control line if optional line driver is use	PU (1)
No Function at Boot. User I/O after Boot	62	GPIO_2	Not defined during Boot Up	PD (0)
No Function at Boot. User I/O after Boot	63	GPIO_3	Not defined during Boot Up	PD (0)

While the QCA6410 has low current internal pull up and pull down resistors it is recommended that all Configuration Straps have external resistors for easy connections and changing during troubleshooting and test.

- GPIO[0] Boot from external NVM (flash) device using the SPI Master bus
- The following two figures show the circuit arrangement for GPIO pull-up and pull- down.

  NOTE These are the connections if a LED is desired on one of these pins.
- Do not connect any GPIO directly to ground or to the 3.3 V bus. After the RESET_L signal goes high FW may drive any GPIO pin to an output. Any pin may be driven high or low by FW to indicate system status using the LED outputs.

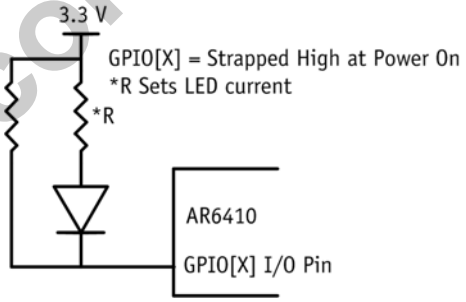


Figure 2-5 QCA6410 Pull-Down LED Strapping

Qualcomm Atheros Confidential

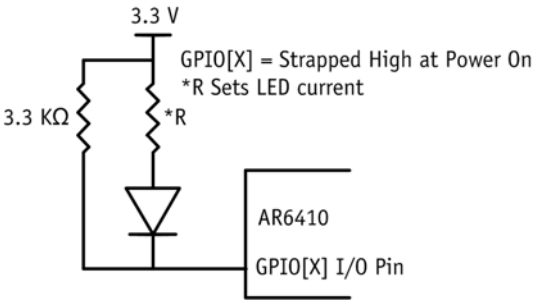


Figure 2-6 QCA6410 Pull-Up LED Strapping

The following additional straps are also available in any mode of operation:

Table 2-4 Special Straps for Test

Symbol	Pin	Strap=1	Strap=0
PLL-Bypass	44	Used for test. 1 = disable the PLL	Normal mode. PD (0) internally

Special test conditions are enabled by the pin straps listed in Table 2-4.

Table 2-5 Other Internal Pull-Up Pull-Down Signal List

Symbol	Pin	Description	Internal Pull-Up/Down
TCK	23	JTAG Test Clock Out	PU (1)
TDI	22	JTAG Test Data In	PU (1)
TMS	25	JTAG Test Mode Select	PU (1)
TRST_L	27	JTAG Test Reset	PU (1)
SERIAL_IO[0]	20	Shared_GPIO[4]	PD (0)*
SERIAL_IO[1]	19	Shared_GPIO[5] or UART RTS	PD (0)*
SERIAL_IO[2]	15	Shared_GPIO[6] or UART CTS	PD (0)*
SERIAL_IO[3]	14	Shared_GPIO[7] or UART Serial Out	PD (0)*
SERIAL_IO[4]	13	Shared_GPIO[8] or UART Serial In	PD (0)*

The Mode of the SERIAL_IO[X] pins is set by an internal register by FW. The Internal Pull-Up/Pull down for these pins applies only when the pin is an input as in Shared_GPIO[X] mode or some SPI Slave bus or UART input function.

The main chip reset is the active low RESET_L input. This reset input is first passed through a Schmidt trigger input, and then goes through a simple digital filter before it actually becomes the on-chip reset. This pin is the only filtered input pin.

While this power-on-reset signal is asserted, all clocks on all flops in the chip are active, and all on-chip registers are reset to a default state. To reduce the peak power used during this period of time, the frequency of all clocks is 8 times lower than normal.

Once RESET_L is de-asserted (goes high), the on-chip reset is further extended by a ~4000 clocks to allow the on-chip PLL to lock. At the end of this delay, the internal clocks are brought up to their normal operating frequency, and then the majority of them are gated off. Once the ARM

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

processor is brought out of reset, the processor jumps to address 0xFFFF0000 and the Boot ROM software takes over.

In addition to these hardware initiated resets, there are also numerous sources of software controlled reset within the QCA6410. A MME command on the PLC bus can reset the entire chip as well as the ARM processor. This option is again selected through a software configuration register.

Internal to the QCA6410, there are also numerous sources of reset. The processor can write to internal registers to take individual modules in or out of reset, as well as initiate an entire chip reset. Furthermore, to recover from system problems, a watchdog reset circuit is available that initiates a reset if the software or system have crashed.

The Shared_GPIO[X] pins are not the same as the GPIO pins described earlier. These signal are only available when enabled on the Serial _IO pins.

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

3 System Architecture

3.1 Architecture Block Diagram

The QCA6410 SoC and MAC baseline architecture is represented in Figure 3-1.

3.2 Host Interfaces

The QCA6410 MAC/PHY Transceiver is designed to bridge multi-stream Ethernet content from a powerline network to an Ethernet 10/100 Mbps 802.3 network. Examples include high and standard definition television (HDTV, SDTV), and other digital multimedia file sharing and data applications. The Powerline Communications (PLC) specific media access controller (MAC) manages network admission and service flows to maximize the quality of service (QoS) over the powerline network.

Other serial and parallel ports provide additional I/O and user LED indicators to support the application.

3.2.1 QCA6410 SPI Master Interfaces

The QCA6410 includes a SPI Master bus to load FW from an external Non Volatile Memory (NVM -Flash) memory. The SPI Master Bus has dedicated pins and is always available to load the QCA6410 application Firmware (FW). The SPI Master interface signals are outlined in Table 3-1 and shown in Figure 3-2. The SPI Master port has dedicated un-multiplexed data, clock and chip-select pins and is always used for FW load. The NVM write and read cycle timing is not programmable and is set by the QCA6410 hardware.

Table 3-1 SPI Master Signal List

Signal Name	Description
SPI_CLK	SPI Master Clock Output
SPI_DO	SPI Master Data Output
SPI_DI	SPI Master Data Input
SPI_CS_L	SPI Master Chip Select Low

Qualcomm Atheros Confidential

3.2.2 QCA6410 Application Interface

A serial UART interface and individual parallel GPIO bits are available for application interface. The UART supports frequencies up to 1MBAUD, and has four signal pins: TX Data, RX Data, RTS and CTS. The use of RTS and CTS is completely optional.

Due to pin limitations, only some interface pins, the UART or the Shared GPIO may be used across the 5 I/O multiplexed pins allocated. These options are outlined in Table 3-2 and are shown in Figure 3-1.

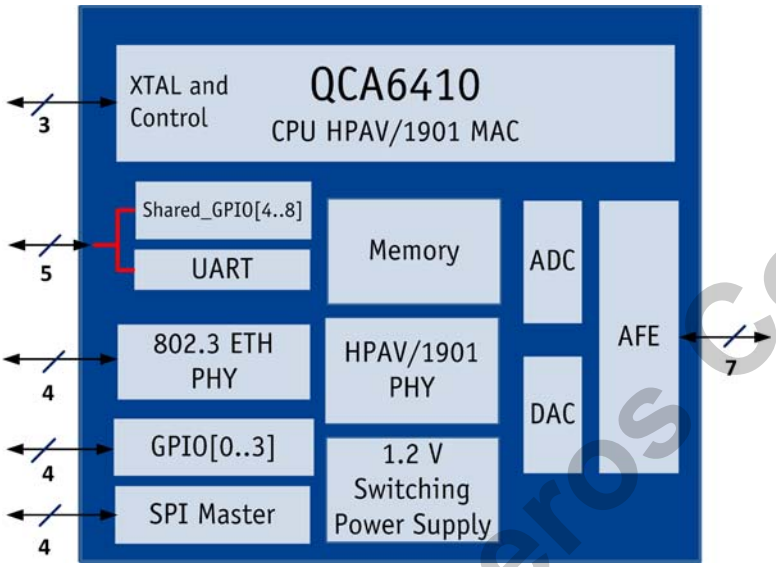


Figure 3-1 QCA6410 SoC and MAC Architecture Block Diagram

Table 3-2 Host Serial Interface Signal List

I/O Pad	GPIO Mode	UART Mode	UART with RTS/CTS Mode
SERIAL_IO[0]	SHARED_GPIO[4]	SHARED_GPIO[4]	SHARED_GPIO[4]
SERIAL_IO[1]	SHARED_GPIO[5]	SHARED_GPIO[5]	RTS
SERIAL_IO[2]	SHARED_GPIO[6]	SHARED_GPIO[6]	CTS
SERIAL_IO[3]	SHARED_GPIO[7]	Serial Out	Serial Out
SERIAL_IO[4]	SHARED_GPIO[8]	Serial In	Serial In

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

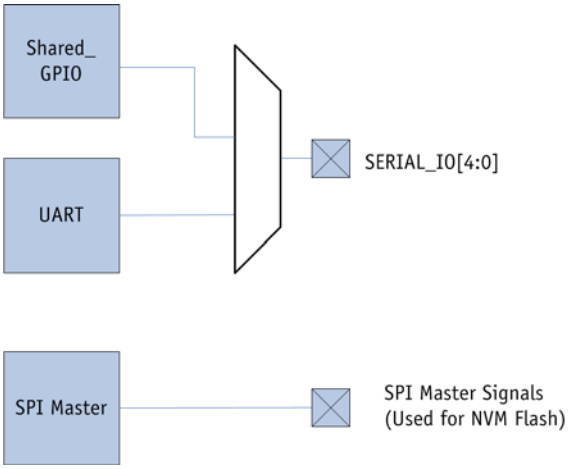


Figure 3-2 Serial I/O Options

3.2.3 GPIO

The QCA6410 supports a maximum of 4 General Purpose I/O signals. Each of these can be individually programmed as an input or output; outputs can be further programmed for different drive-strength for different applications. Up to 2 of these GPIO signals (depending upon configuration) can be used to trigger interrupts to the main ARM processor. Interrupts can be any of active high or low, or positive/negative edge triggered.

In addition to the normal functions associated with the GPIO pins, there are also additional debug functions associated with the GPIO I/O pads. These functions are shown within Table 3-3.

Table 3-3 GPIO Debug MUX Options

Pin	Normal Function	Clock Observability #1	Clock Observability #2
GPIO_0	GPIO_0	25 MHz Oscillator: Positive	25 MHz Oscillator: Positive
GPIO_1	GPIO_1	25 MHz Oscillator: Negative	25 MHz Oscillator: Negative
GPIO_2	GPIO_2	75 MHz PLL: Positive	
GPIO_3	GPIO_3	75 MHz PLL: Negative	

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

4 Functional Description

4.1 Overview

The QCA6410 HomePlug-AV PHY integrated circuit is a PLC MAC/PHY Transceiver with serial data I/O. The device is designed to bridge lower data rate multi-stream PLC based Ethernet content from a HomePlug or HomePlug AV Powerline network to products in the home, office or automotive environment.

The Powerline communications (PLC) specific MAC manages network admission and service flows to maximize the quality of service (QoS) over the HomePlug network.

The single QCA6410 HPAV IC contains many of the functions previously found in higher data rate Atheros two chip PLC system solutions.

4.2 Power Management

4.2.1 Power Domains

To address the requirement that devices consume less power especially when they are not actively being used, the QCA6410 hardware and firmware supports power management functions. Power usage will be controlled using a combination of power gating and clock gating. The system contains two power islands, the Always Powered Island, and the Power Gated Island.

- Always Powered:
  25 MHz Osc to PMUAlways On
- Always Powered, Low Power Mode:
  All Clocks Stopped in Sleep Mode
- Power Gated, Clock Gated:
  All Clocks stopped in Sleep Mode

4.2.2 Power Management Unit

Through a combination of hardware and software, four states of power consumption are supported shown in Table 4-1.

To implement the power states shown in Table 4-1, the QCA6410 includes four Power Management states. Sleep and deep sleep are low power modes to reduce the chipset power consumption significantly from more active power consumption states.

Qualcomm Atheros Confidential

Table 4-1 Supported Power States

Power Mode	Functionality	Wake-up to Active
Deep Sleep	■ Lowest power consumption □ Utilizes internal power gating ■ Not capable of processing Powerline activity ■ Not capable of receiving host wake commands on UART ■ Not capable of monitoring link status ■ Not capable of supporting 802.3.az low power idle/active transitions ■ Transition time to active is a function of the warm boot time after the detection of a wake event	■ Internal timer ■ External GPIO input signal transition
Sleep	■ Low power consumption □ All blocks powered, utilizes clock gating □ HomePlug PHY off, HomePlug MAC on ■ Not capable of processing Powerline activity ■ Capable of receiving host wake commands ■ Transition time to active is a function of the warm boot time after the detection of a wake event ■ > 75% power savings over the Active state	■ Internal timers ■ External GPIO input signal transition
Standby	■ Power consumption is programmable as a function of the sleep schedule; i.e. time in the sleep state or deep sleep state versus the time in the active state □ All blocks powered, utilizes clock gating □ HomePlug PHY on, HomePlug MAC on ■ Capable of processing Powerline messages when the ARM, HomePlug MAC and HomePlug Powerline Phy are active ■ Capable of receiving host wake-up commands always ■ The response time to Powerline wake events is a function of the duty-cycle configuration and firmware recovery time	■ Any MagicPacket including specific values from Powerline ■ Wakeup messages from Powerline ■ Internal timers ■ External signal GPIO input transition ■ Arrival of a broadcast, multicast, or unicast packet from Powerline
Active	■ Highest power consumption under one of three conditions transmit, receive or idle ■ Capable for full operation at maximum bandwidth available based upon the environment ■ Capable of dropping to lower power-state based upon traffic patterns or host messages ■ Transition time to Standby or Idle within 1 s of sleep command or event	

Qualcomm Atheros Confidential

In the sleep state, all QCA6410 logic except the power management unit (PMU), the SRAM containing the primary CPU execution image, some Analog IP and a few I/O cells are put into an inactive but powered state. Software and timers or external events from the GPIO interface are used to wake-up the device.

In the deep sleep state, all QCA6410 logic except the power management unit (PMU), the SRAM containing the primary CPU execution image, some Analog IP and a few I/O cells are powered off by the PMU. Timers or GPIO are used to wake-up the device.

The standby power state is a user-programmable low power state to allow trade-offs between response or transition time from idle to active and the average power consumption in idle mode. The QCA6410 toggles between standby and active based on programmable timers in the PMU and controls the AFE IC accordingly.

4.3 Boot Options

The QCA6410 contains an embedded ARM926 processor that requires Firmware (FW) loaded from an external device or host processor source. FW must be loaded to enable data transfer/processing to and from the Powerline. As this firmware does not reside on-chip, the QCA6410 boot sequence requires the ability to download configuration and code from an external device like a Non Volatile Memory NVM (flash) device using the Master SPI bus.

Straps on the QCA6410 specify which method is used to access FW. All applications of the QCA6410 can boot from flash using the SPI Master Port if NVM is present. Reading NVM Flash (connected through pins SPI_CLK, SPI_DO, SPI_DI, and SPI_CS_L) loads FW immediately after RESET_L goes high. The QCA6410 can boot from an external host device over the Integrated Ethernet port interface using a sequence of low level software instructions residing in the QCA6410 ROM. These are MME command packets.

Once FW is loaded and running in the QCA6410 from any source it begins communication over any and all I/O ports including the Powerline. Powerline commands can program or re-program the local NVM device if available. There are several methods then of updating FW using a sequence of simple MME packets. To boot the chip, an applet is first downloaded to configure the internal hardware. Next, the chip downloads the firmware and then hands CPU control over to this program to start the Powerline network. At this point the Powerline network has full control of the QCA6410.

In a manufacturing environment in which a blank flash resides on the board, the QCA6410 will automatically fail-over to the Integrated Ethernet bus interface to allow the device to be programmed using the Production Test System.

The QCA6410 supports SPI flash NVM devices supporting the ST SPI-flash command-set. See Table 4-2. for qualified NVM memory devices.

Table 4-2 QCA6410 FLASH Recommended Devices

Vendor	Part Number	Speed Grade	Memory Size	Comments
Atmel	AT26DF161-SU	100 MHz	16-Mbits	[1]Not for new designs
Atmel	AT26DF161A-SU	70 MHz	16-Mbits	[1]

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 4-2 QCA6410 FLASH Recommended Devices

Vendor	Part Number	Speed Grade	Memory Size	Comments
Numonyx	M25P80-VMW6TP	75 MHz	8-Mbits	[1]
Numonyx	M25P16-VMW6TP	75 MHz	16-Mbits	[1]
Macronix	MX25L8005	70 MHz	8-Mbits	[1]
Macronix	MX25L1605	86 MHz	16-Mbits	[1]

1. Verified for use with the QCA6410

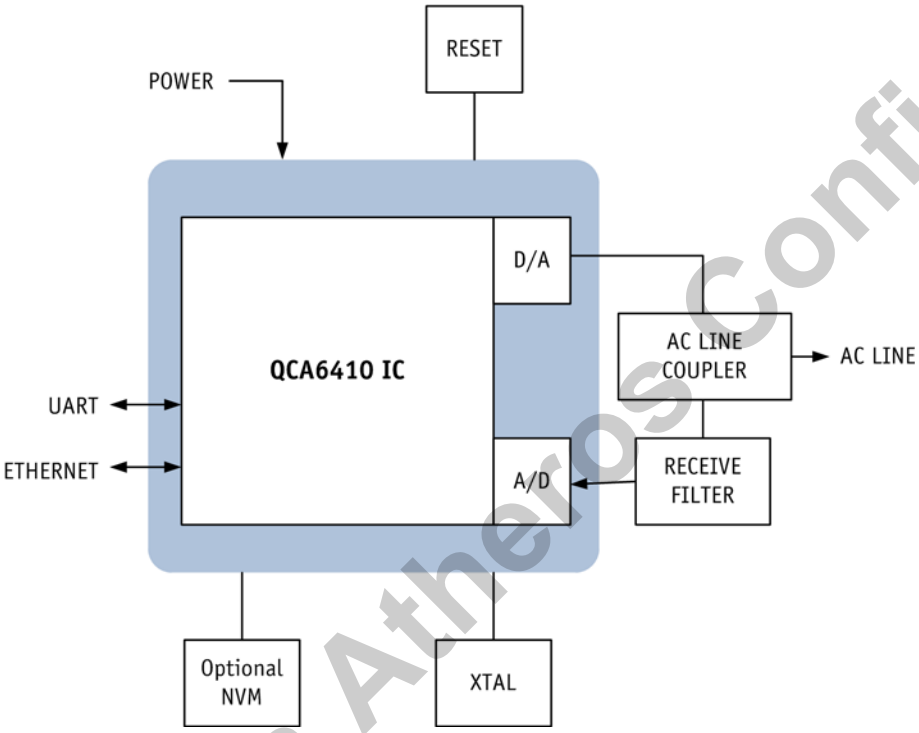


Figure 4-1 QCA6410 Powerline Station

4.4 Serial Interfaces

The QCA6410 includes multiple serial interfaces:

- UART
- GPIO
- SPI Master (SPI) for NVM -flash

The UART supports four signal pins for various applications: TX Data, RX Data, Request to Send (RTS) and Clear to Send (CTS). The use of RTS and CTS is optional. The UART does not support any Infra-red Data Association (IrDA) functionality.

The General Purpose Input/Output (GPIO) is available for use for discrete sensing and control.

The UART port shares its I/O with the Shared_GPIOs. Their operation is mutually exclusive.

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

The Serial Peripheral Interface (SPI) master port has one chip select, just for boot from NVM-flash. The SPI master port is always available and supports boot from NVM-Flash.

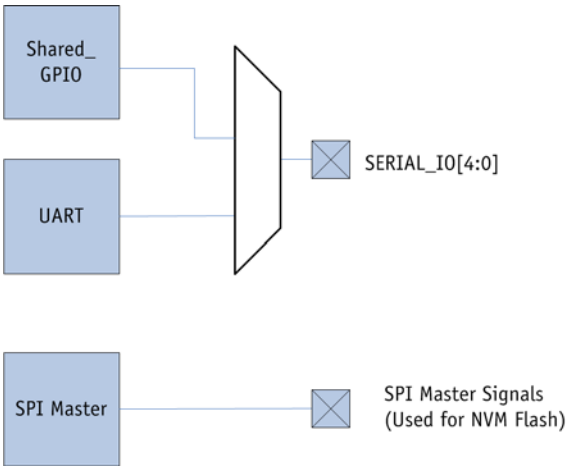


Figure 4-2 Serial I/O Options

CPU oversees operation of the integrated HPAV PHY via an interface that carries control/status and transmits and receives data packets.

General-purpose I/O pins are available to drive light emitting diodes (LED) to indicate status, user reset, and network-attach events. A dedicated sub-system is responsible for power-management functions on the QCA6410. Two on-board phase locked loops (PLL) and a built-in crystal oscillator cell are used to generate the required system clocks from a single external 25 MHz crystal. Various loop-back modes support manufacturing test.

Table 4-3 Peripheral Pin-out Options

Pin	Symbol	Configuration	
		UART	Shared_GPIO
67	SPI_CLK	SPI Master Clock Output	
66	SPI_DO	SPI Master Data Output	
3	SPI_DI	SPI Master Data Input	
65	SPI_CS_L	SPI Master Chip Select #1	
20	SERIAL_IO[0]	Shared_GPIO[4]	Shared_GPIO[4]
19	SERIAL_IO[1]	RTS or Shared_GPIO[5]	Shared_GPIO[5]
16	SERIAL_IO[2]	CTS or Shared_GPIO[6]	Shared_GPIO[6]
15	SERIAL_IO[3]	TXD	Shared_GPIO[7]
14	SERIAL_IO[4]	RXD	Shared_GPIO[8]

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

5 Electrical Characteristics

5.1 Absolute Maximum Ratings

Table 5-1 summarizes the absolute maximum ratings and Table 5-2 lists the recommended operating conditions for the QCA6410.

Absolute maximum ratings are those values beyond which damage to the device can occur. Functional operation under these conditions, or at any other condition beyond those indicated in the operational sections of this document, is not recommended

Table 5-1 Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Units
Electrical				
VCORE	Core Supply Voltage	-0.3	1.32	V
VPLL_1	PLL Analog Power	-0.3	1.32	V
VDDIO	I/O Supply Voltage	-0.3	3.63	V
VPLL_3	Analog PLL Supply Voltage	-0.3	3.63	V
AVDD_1	Analog Power 1 V	-0.3	1.32	V
AVDD_3	Analog Power 3 V	-0.3	3.63	V
OSCx, PLLx	PLL lines	-0.3	VPLL_1+0.3	V
ADCx, DACx	Analog lines	-0.3	AVDD_1+0.3	V
T_{store}	Storage Temperature	-40	+150	ºC
T_j	Junction Temperature	—	+125	ºC
Digital	All other digital lines	-0.3	VDDIO+0.3	V
ESD Immunity	Human Body Model	—	2.0	kV

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

5.2 Recommended Operating Conditions

Table 5-2 Recommended Operating Conditions [1]

Symbol	Parameter	Min	Typ	Max	Units
VCORE	Core Supply Voltage	1.14	1.2	1.26	V
VDDIO	QCA6410 I/O Supply Voltage	3.13	3.3	3.46	V
AVDD_1	Analog Power 1.2 V	1.14	1.2	1.26	V
AVDD_3	Analog Power 3.3 V	3.13	3.3	3.46	V
Tcase[2]	Top of Case Temperature	—	—	115	ºC
Psi_{JT}[2]	Thermal Parameter	—	—	2.8	ºC/W

1. Power supply voltages may be applied or removed in any sequence during power up or power down.
2. For the QCA6410 8x8 mm 68-Pin QFN package the IC will operate at its respective Tcase temperature range provided required PCB footprint and thermal design layout is used. See Qualcomm Atheros reference design board layout.

5.3 QCA6410 Power Supply Requirements

Table 5-3 QCA6410 Typical Power Supply Requirements[1]

Symbol	Parameter	Supply Voltage (V)	Transmit Mode Current (mA)	Receive Mode Current (mA)
VCORE	QCA6410 Core Supply Voltage	1.2	245	312
VDDIO	Total for VDDIO, VDDIO_E and VDDIO_D	3.3	547	220

1. Data in Table 5-3 is typical.
   ASSUMPTION: QCA6410 PL22 Hardware Reference Design.

Table 5-4 QCA6410 Typical Total IC Power

Device	Transmit Mode Power (mW)	Receive Mode Power (mW)
QCA6410	2100	1100

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

5.4 DC Switching Thresholds

Table 5-5 DC Switching Thresholds

Symbol	Parameter	Test Conditions	Min	Max	Units
V_{IL}	Low-level input voltage		—	0.8	V
V_{IH}	High-level input voltage		2.0	—	V
V_{OL}	Low-level output voltage	I_{OL}= 4 mA, 12 mA[1]	—	0.4	V
V_{OH}	High-level output voltage	I_{OH} = -4 mA, -12 mA[2]	2.4	—	V
I_{IL}	Low-level input current	V_I = Gnd	-1	—	µA
I_{IH}	High-level input current	V_I = VDDIO	—	1	µA
I_{OZ}	High-impedance output current	Gnd ≤ VI ≥ VDDIO	-1	+1	µA

1. IOL = 12 mA for all GPIOs
   IOL = 4 mA for all other interfaces
2. IOH = -12 mA for all GPIOs.
   IOH = -4 mA for all other interfaces

5.5 Crystal Specifications

Table 5-6 Crystal Specifications

Parameters	Values	
Frequency	25 MHz	
Frequency Tolerance @ +25°C	±10 ppm Max.	
Frequency Stability: 0°C to +70°C	±10 ppm Max.	
Crystal Cut	AT Strip	
Effective Series Resistance	15 Ω Typ., 50 Ω Max.	
Shunt Capacitance	7 pF Max.	
Mode of Operation	Fundamental	
Aging	±3 ppm/year Max.	
Storage Temperature	-40°C to +85°C	
Operating Temperature	0°C to +70°C	
Package Type	HC-49/U	SMD
Recommended Parts	ECS-250-1B-4X-F-C	7M25000013
Load Capacitance (CL)	18 pF	10 pF
Recommended Drive Level	100 µW Typ.	100 µW Typ., 200 µW Max.

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

6 Timing Constraints

6.1 System Clock and Reset Timing

6.1.1 QCA6410 Crystal Oscillator

The QCA6410 IC requires a 25 MHz Clock Signal for a master time base. All timing is derived from this signal. This signal is normally derived from a 25 MHz crystal attached to the XTAL_IN and XTAL_OUT pins 37 and 38 respectively. See schematic 60006410 for details. The two trim capacitors (C185 and C286) are nominally 4.7 pF but these values are dependent on part spacing and layout details that vary depending on board design. The designer must measure the master oscillator frequency and adjust these two trim capacitors to deliver the correct frequency for any new board design. Accuracy of the master oscillator is essential to correct operation of all PLC designs. See Section 6.1.2 for accuracy details. An external 25 MHz Clock Signal can be used rather than the integrated oscillator.

Table 6-1 System Clock and Reset Timing Specifications

Parameter		Min	Typ	Max	Units
25.0 MHz Oscillator					
t_{CLKp}	OSCIN Period	40.000 - 23 ppm	—	40.000 + 23 ppm	ns
Tr/Tf	Rise/Fall Time (20% to 80%)	—	—	1	ns
Duty_CLK	Duty Cycle for OSIN	45	50	55	%
tosc	Oscillator Start-Up Time	—	0.75	TBD	ms
Internal Reset					
t_{RSTa}[1]	Internal Reset Time		4000		Clock Cycles
GPIO Configuration and Reset Timing					
t_{GPIO_SETUP}	GPIO Config. SETUP	TBD	—	—	ns
t_{GPIO_HOLD}	GPIO Config. HOLD	TBD	—	—	ns
t_{RSTa}	RESET_L	—	—	—	µs

1. Internal reset is active after power is applied and goes inactive after 4000 clock cycles if both DVDD and VDD voltages meet minimum specifications. See marker (C) in the timing see Figure 6-1.

Qualcomm Atheros Confidential

6.1.2 Master Time Base Accuracy

The accuracy of this signal must meet the HomePlug standard. Initial accuracy as the QCA6410 is powered-on must be ±10 ppm at specified operating voltage. Additionally the 25 MHz signal may drift an additional ±10 ppm over the operating temperature range. In addition the frequency may vary ±3 ppm per year to account for aging of the crystal. See Table 5-6 and Table 6-1 for details.

6.1.3 External Clock Input

A nominal 1.1 Vpp clock signal may be applied to XTAL_OUT (Pin 38). The XTAL_IN (Pin 37) must be pulled to ground through a zero Ω resistor for proper QCA6410 circuit bias and to turn off the internal oscillator circuit. See schematic 60006410 for details and always use the latest Qualcomm Atheros documents for component values and circuit arrangements. OSCIN is a critical signal. To achieve optimum system performance there must be no deviation from the recommend circuit.

Two on-board PLLs use the Master Time Base signal to generate the required internal QCA6410 system clocks. See the PL22 Reference Design schematic 60005310 and PCB design file 90005310 for details.

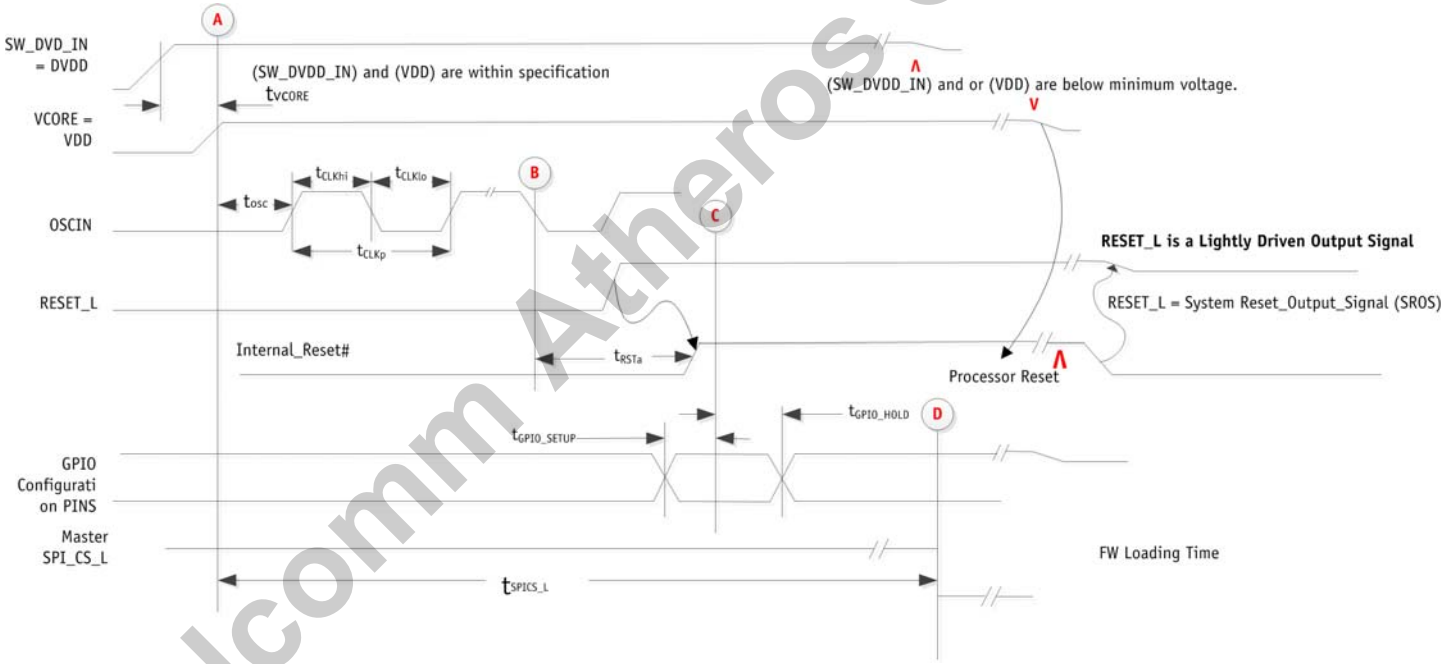


Figure 6-1 System Clock and Reset Timing

The RESET_L pin is an input signal and a lightly driven output. There is no need to drive the RESET_L pin in most standalone applications. If there is no external driver on the RESET_L pin it can be monitored by an oscilloscope to determine when the QCA6410 is in reset. See the PL22 Reference Design schematic 60005310 and PCB design file 90005310 for details.

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

6.2 Zero Crossing Detector Timing

Zero Crossing is internal to the QCA6410 IC.

Table 6-2 Zero Crossing Detector Timing Specifications

Symbol	Parameter	Min	Typ	Max	Units
T_{LOW}	Active Low time	400	1/(2F)	1/F-T_{LOW}	µs
T_{HIGH}	Active High Time	400	1/(2F)	1/F-T_{HIGH}	µs
T_R	Rise time	1	—	1,000	µs
T_F	Fall time	1	—	1,000	µs
T_{OFFSET}	Offset from zero crossing	0	—	1/F	
F_{60}	60 Hz frequency tolerance	1	[1]	[1]	%
F_{50}	50 Hz frequency tolerance	[1]	[1]	[1]	%

1. Frequency accuracy determination is a firmware function. Consult the applicable firmware technical reference manual for this information.

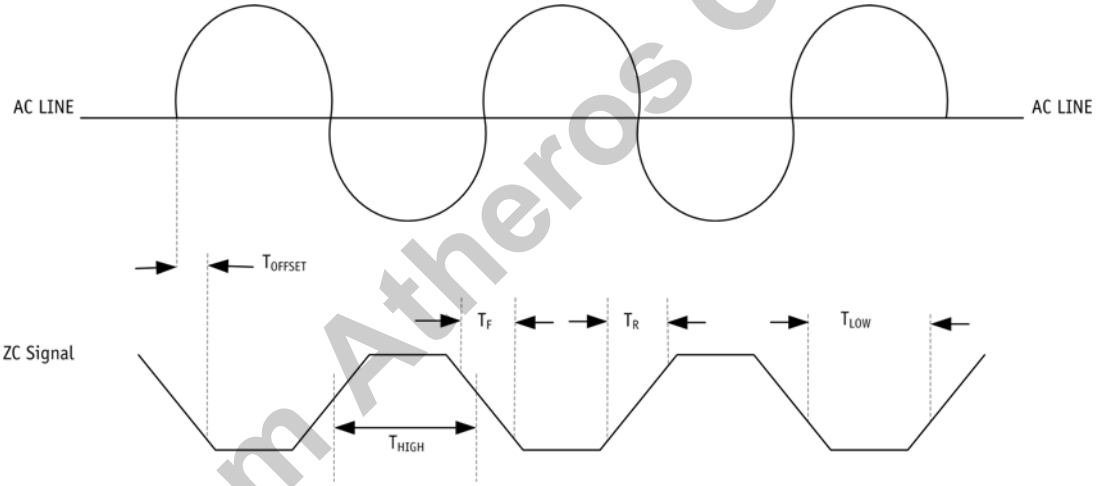


Figure 6-2 ZC_IN Timing

6.3 ISPI (Boot FLASH) Timing

Table 6-3 SPI Timing Specifications

Parameter		Min	Max	Units
t_{CLKp}	SPI_CLK Period	25	—	ns
t_{CLKhi}	SPI_CLK High Time	11	—	ns
t_{CLKlo}	SPI_CLK Low Time	11	—	ns
t_{OV}	Output Valid Time	5	—	ns
t_{OH}	Output Hold Time	5	—	ns

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 6-3 SPI Timing Specifications

Parameter		Min	Max	Units
t_{SU}	Input Setup Time	5	—	ns
t_H	Input Hold Time	5	—	ns

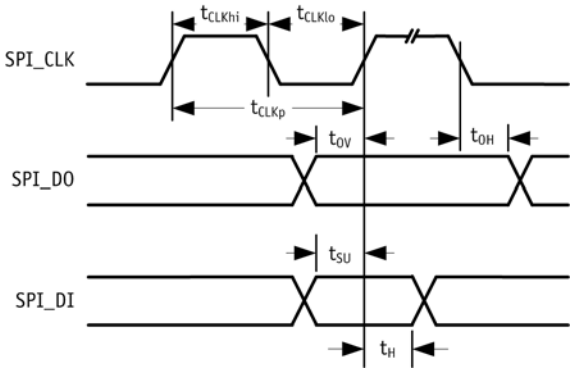


Figure 6-3 SPI Timing Diagram

Qualcomm Atheros Confidential

7 Package Dimensions

The QCA6410 is packaged in a 68-pin QFN package. The body size is 8 mm by 8 mm. The package drawings and dimensions are provided in Figure 7-1, Table 7-1 and Figure 7-2, Table 7-2.

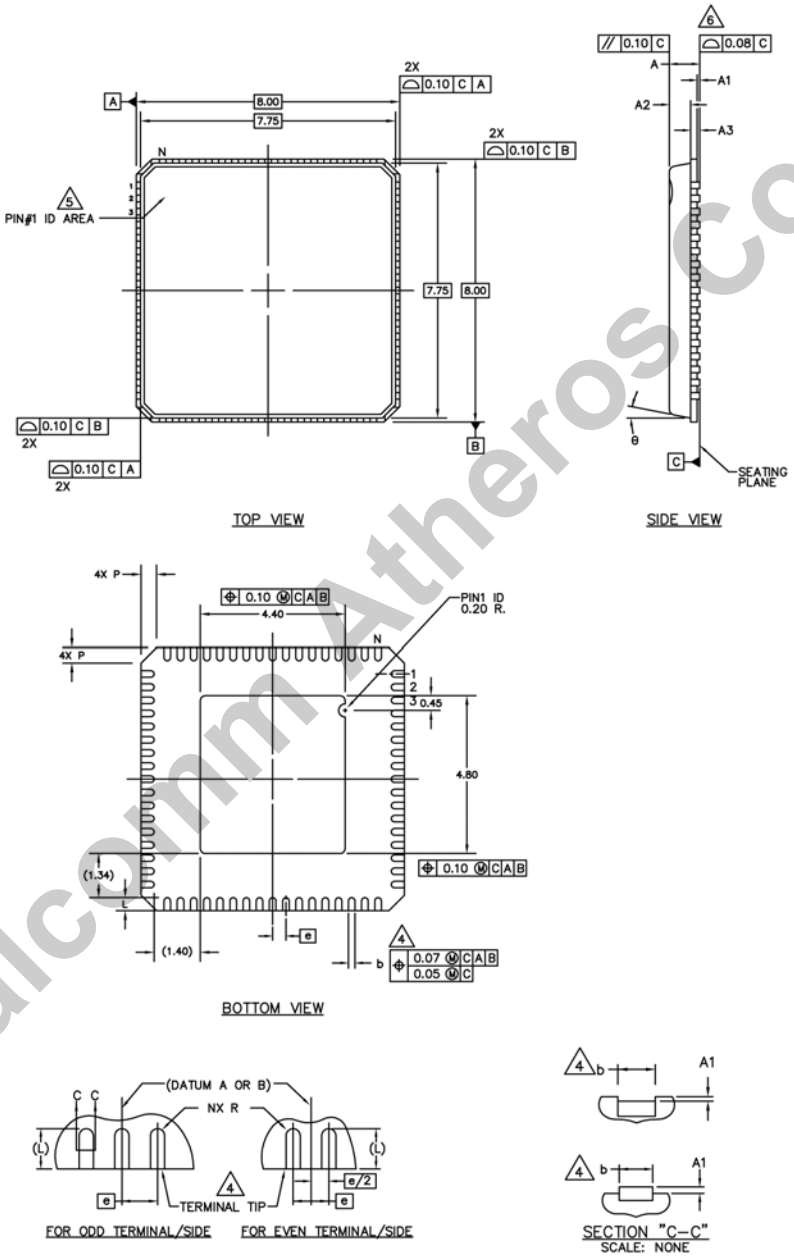


Figure 7-1 Detailed QCA6410 QFN1 Package Dimensions

Qualcomm Atheros Confidential

Table 7-1 Common Dimensions QFN1 [8]

Symbol	Min.	Nom.	Max.	Note
A	0.80	0.85	0.90	
A1	0.00	0.01	0.05	[6]
A2	0.60	0.65	0.70	
A3		0.20 REF.		
q	0	—	12º	
P	0.24	0.42	0.60	
e		0.40 BSC		
N		68		[3]
L	0.30	0.40	0.50	
b	0.15	0.20	0.25	

NOTES:

[1]. Die thickness allowable is 0.305 mm maximum (0.012 inches maximum)

[2]. Dimensioning and tolerances conform to ASME Y14.5M. - 1994.

[3]. N is the number of terminals

[4]. Dimension b applies to plated terminal and is measured between 0.15 mm and 0.30 mm from terminal tip.

[5]. The pin #1 identifier must be existed on the top surface of the package by using indentation mark or other feature of package body.

[6]. Unilateral coplanarity zone applies to the exposed pad as well as the terminals.

[7]. Package warpage max 0.08 mm.

[8]. All dimensions are in millimeters.

Qualcomm Atheros Confidential

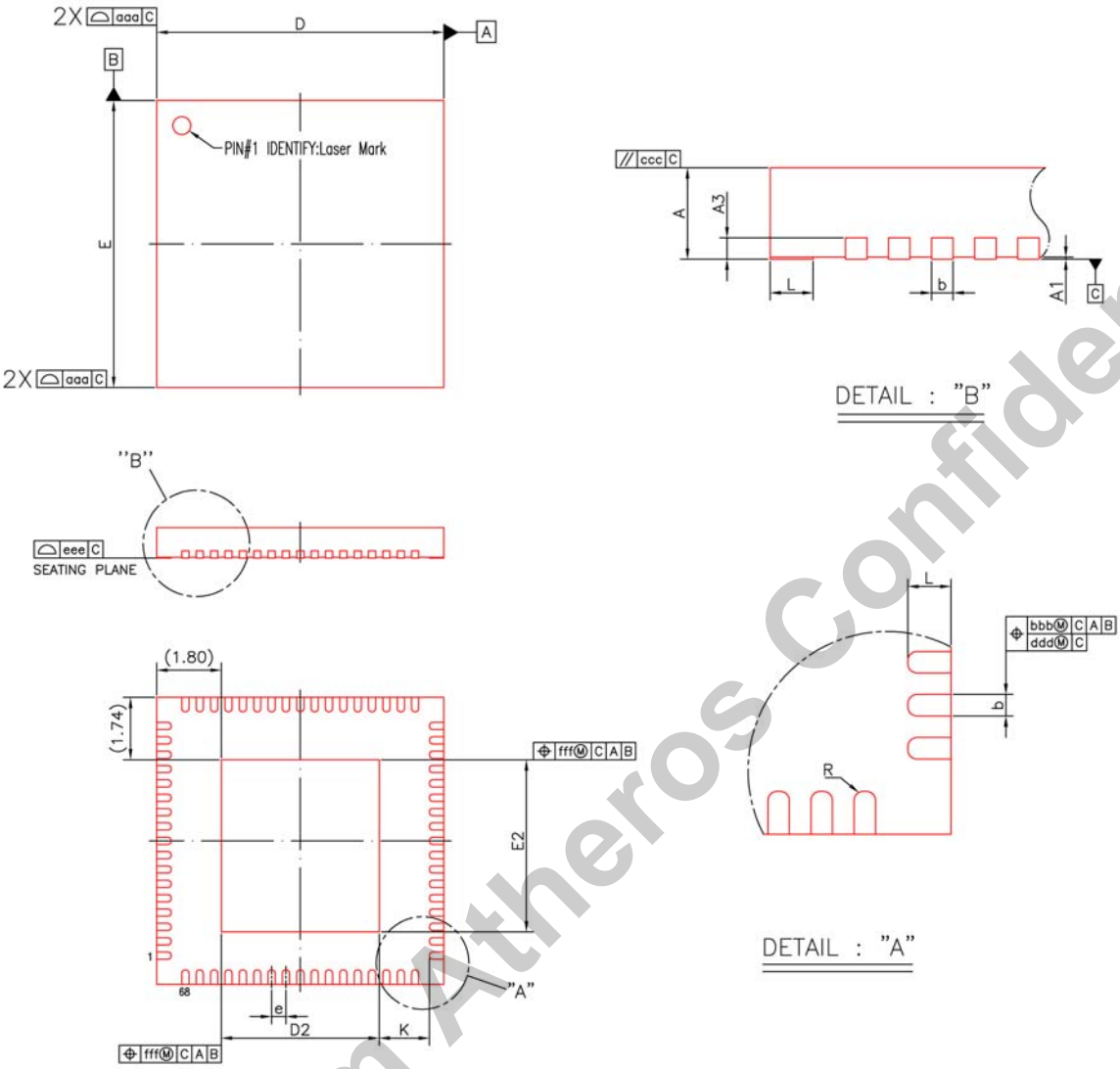


Figure 7-2 Detailed QCA6410 QFN2 Package Dimensions

Qualcomm Atheros Confidential

MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Table 7-2 Common Dimensions QFN2

Symbol	Min.	Nom.	Max.
A	0.80	0.85	0.90
A1	0.00	0.02	0.05
A3		0.20 REF.	
b	0.15	0.20	0.25
D/E	7.90	8.00	8.10
D2	4.25	4.40	4.55
E2	4.65	4.80	4.95
e		0.40 BSC	
L	0.30	0.40	0.50
R	0.075	—	—
K	0.20	—	—
aaa		0.10	
bbb		0.07	
ccc		0.10	
ddd		0.05	
eee		0.08	
fff		0.10	

NOTES:

[1]. Controlling dimension: millimeters.

[2]. Reference document: JEDEC MO-220.

Qualcomm Atheros Confidential

8 Ordering Information

The order number QCA6410-AL3C specifies a lead-free standard-temperature version of the QCA6410.

The order number QCA6410-AL3C-R specifies a lead-free standard-temperature version of the QCA6410 in tape and reel.

Qualcomm Atheros Confidential